RED SEAL EXAM PREP

Gasfitter (Class A)

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Grouped by this site's own topic areas
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Mock Exam — answer all questions, then check the key

Q1. **A service regulator with a 60 psig inlet is set for 7 in WC outlet. With every appliance off, the lockup pressure at the test port reads 14 in WC. What decides whether this reading is acceptable?**

A) Whether it stays within ten percent of the 7 in WC outlet setpoint

B) Nothing — lockup is only valid when read with an appliance firing

C) Whether it stays within the downstream piping and appliance limits

D) Nothing — any lockup above the outlet setpoint condemns a regulator

Q2. **What is the difference between primary air and secondary air in gas burner combustion?**

A) Primary air comes from outside; secondary air is recycled flue gas

B) They are the same air entering from different locations

C) Primary air is for ignition only; secondary air maintains the flame

D) Primary air premixes with gas; secondary air surrounds the flame

Q3. **What unit is typically used to measure gas pressure in low-pressure residential systems?**

A) psi (pounds per square inch)

B) Pascals, the SI pressure unit

C) Bar (metric pressure unit)

D) Inches of water column (in WC)

Q4. **Which of the following is acceptable as the manual shut-off valve serving an individual gas appliance?**

A) A ball or lubricated-plug valve certified for gas service

B) A solenoid valve wired into the appliance's own control circuit

C) A valve rated for the line pressure, whether or not it is certified for gas

D) A gate valve, so the input can be throttled at the valve

Q5. **Where should a combustible gas detector be positioned for a natural gas leak vs. a propane leak?**

A) Natural gas: near the ceiling; propane: near the floor

B) Always at floor level for both natural gas and propane

C) Detector position does not matter — both gases disperse evenly

D) Always at breathing height (1.2 m) for both gases

Q6. **Why must every joint in a positive-pressure appliance vent be sealed and mechanically fastened?**

A) Because sealed joints let the vent be one size smaller than the tables

B) Because the flue gas is above room pressure, so a leak enters the room

C) Because unsealed joints cool the vent below the flue gas dew point

D) Because the sealant is what carries the vent load between supports

Q7. **What is the purpose of a dip tube in a gas storage water heater?**

A) To protect the anode rod from sediment build-up

B) To connect the T&P relief valve to the drain

C) To measure the water level inside the tank

D) To direct cold inlet water to the tank bottom

Q8. **Steel gas piping passes through a poured concrete foundation wall. Under CSA B149.1, what is required for the portion of pipe that runs through the concrete?**

A) Wrap it in fibreglass insulation to isolate it from the wall

B) Sleeve it, or double-wrap it with pipe wrap tape

C) Coat it with epoxy paint before the concrete is poured

D) Cast it directly in the concrete, which shields steel like rebar

Q9. **A gas furnace is to be installed at a site about 1,000 m above sea level. What determines how the appliance is set up for that elevation?**

A) The gas utility, which lowers the pressure it delivers to suit the elevation

B) Its high-altitude certification and the manufacturer's instructions

C) Nothing extra, because the appliance regulator compensates for thin air

D) A larger burner orifice, because the thinner air lets it accept more gas

Q10. **The flames on an atmospheric burner lift off the ports and float above them. What is the most likely cause?**

A) Too much primary air, or gas pressure above the rating

B) Manifold pressure set below the rating plate value

C) Too little primary air reaching the burner mixing tube

D) A partly plugged main burner orifice restricting gas

Q11. **A condensing boiler is installed with a return water temperature of 38°C (100°F). What is the implication of this return temperature?**

A) The boiler is inefficient — return temperature must be above 60°C

B) It enables condensing operation, achieving the highest efficiency

C) Return temperature is too low — the boiler will short-cycle

D) Low return temperature causes carbonic acid corrosion of the heat exchanger

Q12. **An unvented infrared heater is proposed for a truck repair bay. The installation code allows the heater there only if the space is provided with mechanical ventilation sized on the heater's input. What does the code additionally require of that ventilation system itself?**

A) It must be sized to deliver six complete air changes an hour to the repair bay

B) It must be interlocked to a carbon monoxide alarm mounted in the repair bay

C) It must be interlocked so a drop in the required airflow shuts the heater down

D) It must run continuously through every hour the building is occupied by staff

Q13. **What must a Gasfitter Class A gas fitter verify about vent termination height above a flat roof?**

A) A minimum height above the roof surface per CSA B149.1

B) 150 mm above the roof is sufficient for all appliances

C) Vent must extend to the highest point of the building

D) Vent can terminate at roof level — no height requirement

Q14. **Under CSA B149.1, how may copper tubing be used in a propane system?**

A) Type K in liquid service; Type L is restricted to vapour

B) Types K and L, for liquid propane or propane vapour

C) For natural gas alone; propane odourant attacks copper

D) For vapour service alone, after the second-stage regulator

Q15. **A gas-fired storage water heater serving a dwelling unit is set to store water at 60 degrees C. Under the National Plumbing Code of Canada, what does that mean for the bathroom it supplies?**

A) Nothing further is required, as 60 C is the code setting

B) Tub and shower water must be limited to 49 C at the outlet

C) The tank must be turned down, since 60 C is above the cap

D) Tub and shower water must be limited to 43 C at the outlet

Q16. **A propane container is being filled. The filler opens the fixed maximum liquid level gauge and stops the transfer at the moment the white haze leaving that gauge turns to a liquid spray. What has that moment established?**

A) That the container has reached its rated working pressure and is now fully charged

B) That the container relief valve has reseated and is holding after the transfer

C) That the liquid has reached the maximum permitted filling level in the container

D) That air has been cleared from the vapour space above the liquid in the container

Q17. **What is the purpose of an inducer (induced draft) motor on a mid-efficiency or high-efficiency gas furnace?**

A) To circulate conditioned air through the building duct system

B) To pull combustion gases through the heat exchanger to the flue

C) To supply combustion air to the burner before ignition

D) To induce a draft in the chimney by heating the flue pipe

Q18. **An appliance vent is to terminate through a sidewall that faces a paved public sidewalk running along the front of the building. What minimum height above that sidewalk does the installation code require for the vent termination?**

A) 3.0 m (10 ft) above the paved sidewalk

B) 1.8 m (6 ft) above the paved sidewalk

C) 2.1 m (7 ft) above the paved sidewalk

D) 2.4 m (8 ft) above the paved sidewalk

Q19. **What are the four categories of gas appliance venting classification?**

A) Categories I-IV, by the type of gas it burns

B) Categories I-IV, by the vent pipe material used

C) Categories I-IV, by vent pressure and flue loss

D) Categories I-IV, by the appliance input rating

Q20. **A gas fitter calculates that the gas supply pressure at a building 300 m from the main drops to 4 in WC during peak demand. The appliances require a minimum inlet pressure of 5 in WC. What is the solution?**

A) Increase the appliance orifice size to compensate for the low pressure

B) Accept 4 in WC — most appliances tolerate reduced supply pressure

C) Install a booster regulator at the building to raise the delivered pressure

D) Upsize the service pipe or request higher supply pressure from the utility

Q21. **What is the Lower Explosive Limit (LEL) of natural gas and what does it mean?**

A) 15% natural gas concentration — the maximum explosive level

B) 0.5% — the threshold for immediate evacuation

C) 1% natural gas concentration — below this level gas will not ignite

D) 5% gas in air — the minimum concentration that can ignite

Q22. **Complete combustion of natural gas needs a theoretical quantity of air, before any excess air or dilution air is counted. Roughly how much air is that, per volume of gas burned?**

A) About 10 volumes of air for each volume of gas burned

B) About 25 volumes of air for each volume of gas burned

C) About 2 volumes of air for each volume of gas burned

D) About 15 volumes of air for each volume of gas burned

Q23. **What thread sealant must be used on gas pipe NPT connections?**

A) Standard white PTFE tape (plumber's tape)

B) No sealant needed for tapered NPT threads — they self-seal

C) Loctite thread sealant (any grade)

D) Yellow PTFE tape or gas-rated pipe thread compound

Q24. **What is the most common pipe material for above-ground natural gas distribution within buildings?**

A) Type L copper tubing

B) Black steel pipe

C) Schedule 40 PVC plastic pipe

D) Galvanized steel pipe

Q25. **To confirm that a furnace is firing at its rated input, a gas fitter clocks the gas meter with every other gas appliance shut off. What does that test measure directly?**

A) The efficiency of the appliance, once the flue gas temperature is deducted

B) The manifold pressure that the appliance regulator is holding at the burner

C) The volume of gas the appliance draws over a measured length of running time

D) The heating value of the gas the utility is delivering to that meter set

Q26. **What is the purpose of a drip leg (sediment trap) installed before a gas appliance?**

A) To serve as an emergency shutoff in case of fire

B) To allow pressure testing of the appliance separately

C) To trap condensate, debris, and scale from the gas supply

D) To reduce gas pressure to the appliance

Q27. **At what flue gas temperature will condensation occur in a natural gas appliance vent, and why does this matter?**

A) There is no specific dew point — condensation depends only on ambient temperature

B) About 55-60°C (130-140°F) — below this, acidic condensate forms

C) Condensation only occurs in propane appliances — natural gas flue gas does not condense

D) Condensation occurs at 100°C (212°F) — the boiling point of water

Q28. **What is a spillage test for a natural draft gas appliance and when must it be performed?**

A) A check that flue gases go up the vent, not into the room

B) A test for CO in the flue gas after combustion analysis

C) A pressure test for the vent pipe joints

D) A test to detect gas leaking from vent connections

Q29. **A non-recirculating direct gas-fired industrial air heater releases all of the products of combustion from its burner into the air stream it delivers. Under the natural gas and propane installation code, which area may a heater of that kind not supply air to?**

A) A loading dock whose overhead doors are opened through the winter

B) An area of the building in which sleeping accommodation is provided

C) A machine shop that is worked on three shifts around the clock

D) A high-bay warehouse where goods are stored on open steel racking

Q30. **What is the standard low-pressure natural gas outlet pressure from a service regulator to a residential building?**

A) 3.5 in WC (0.87 kPa)

B) 7 in WC (1.75 kPa)

C) 2 psig (13.8 kPa)

D) 14 in WC (3.5 kPa)

Q31. **What is the difference between an intermittent pilot (IPI) and a direct spark ignition (DSI) system?**

A) IPI sparks a pilot first; DSI sparks the main burner directly

B) IPI is for high-efficiency furnaces only; DSI is for boilers only

C) They are identical ignition systems with different names

D) IPI uses a standing pilot; DSI uses a hot surface ignitor

Q32. **What does CSA B149.1 govern?**

A) Gas equipment manufacturing standards

B) The Natural Gas and Propane Installation Code

C) Occupational health and safety for gas fitters

D) Electrical wiring near gas appliances

Q33. **At a winter service call the service regulator at a meter set is found buried in packed snow and ice, with its vent outlet covered. Why must that be cleared before the set is left in service?**

A) The vent is a rain drain, so a blockage causes rust in the spring case but nothing worse

B) The vent must stay open for the regulator to control outlet pressure and relieve safely

C) The vent is the breather for the meter index, so a blockage only stops it registering

D) The vent supplies the air the regulator blends into the gas, so appliances will burn rich

Q34. **The only workable location for a power-vented appliance's sidewall terminal is on the wall beneath an existing wooden deck. The space under the deck floor is closed in with lattice skirting on all four sides, and the top of the terminal would sit 400 mm below the deck joists. How does the installation code treat this location?**

A) Acceptable, provided a listed screened vent terminal cap is fitted to it

B) Acceptable, provided a certified elbow redirects the plume out from under

C) Acceptable, since the 400 mm clearance exceeds the 300 mm minimum given

D) Unacceptable until the space below the deck is fully open on two sides

Q35. **What is second-stage regulation in a propane (LP gas) system?**

A) The regulator reducing first-stage output to appliance pressure

B) The second regulator in a two-stage natural gas system at the meter

C) A backup regulator that activates if the primary fails

D) The second stage of pressure testing for LP systems

Q36. **What are the primary components of natural gas as distributed in Canada?**

A) Primarily methane (CH4), typically 90-95%

B) Equal parts methane, propane, and butane

C) Equal parts hydrogen and methane

D) Primarily propane with methane added for odour

Q37. **What is the FIRST action when you suspect a gas leak in a building?**

A) Open all windows to ventilate and then search for the leak

B) Call 911 from inside the building first

C) Leave the building without operating any electrical switches

D) Attempt to find the leak with a soap solution

Q38. **What is stoichiometric combustion?**

A) Combustion with excess air — the safest type for gas appliances

B) Combustion at the highest possible flame temperature

C) Combustion with exactly the theoretical air required

D) Combustion with insufficient air, producing maximum heat output

Q39. **A gas dryer venting system must terminate where?**

A) Into a laundry room if it has a window

B) Into an approved lint trap inside the building

C) To the exterior of the building only

D) Into the attic space to reduce moisture in the house

Q40. **What is a hot surface ignitor (HSI) and how does it work in a modern gas furnace?**

A) A pilot burner that remains lit continuously between heating cycles

B) A spark ignitor that creates a high-voltage arc to light the burner

C) A UV flame sensor that monitors the main burner flame

D) An element that glows red-hot to ignite the gas burner on contact

Q41. **Tubing is being run inside a hollow stud wall. Under CSA B149.1, what determines whether it must be protected at the studs, joists and plates with No. 16 USG (1.59 mm) steel plates or sleeves?**

A) Whether it is CSST; copper tube is protected by its own wall

B) Whether it lies within 1.75 in (43 mm) of the wall surface

C) Whether the run inside the cavity is longer than 6 ft (2 m)

D) Whether the operating pressure in it is above 14 in WC (3.5 kPa)

Q42. **A sidewall vent terminal serves an appliance with an input of 60,000 BTU/h (18 kW). What is the minimum clearance from the vent termination to a window that can be opened?**

A) 1.8 m (6 ft)

B) 150 mm (6 in)

C) 300 mm (12 in)

D) 900 mm (3 ft)

Q43. **On a large commercial gas boiler, the burner management system runs the combustion air fan for a timed period before the ignition trial is allowed to begin. What sets the required length of that pre-purge?**

A) Chamber and flue volume, and the number of air changes required

B) The warm-up time the hot surface ignitor needs

C) The delay the flame safeguard needs before it can prove the pilot

D) The time the refractory needs to warm up and avoid thermal shock

Q44. **A gas appliance rating plate states "Input: 100,000 BTU/h, AFUE: 80%". What is the useful heat output?**

A) 100,000 BTU/h — all input is usable heat

B) 120,000 BTU/h — efficiency multiplies the input

C) 60,000 BTU/h — safety factor reduces output

D) 80,000 BTU/h — 80% of input is converted to useful heat

Q45. **A direct-vent (sealed combustion) gas appliance draws combustion air from where?**

A) From the building interior through a grille on the appliance

B) From the exterior through a dedicated sealed air pipe

C) From a combination of interior and exterior air

D) From the furnace room through a draft hood

Q46. **How must the horizontal vent connector of a natural draft gas appliance be run between the appliance and the chimney?**

A) Pitch does not matter, since the draft creates its own flow

B) It must be run dead level so that the draft is not disturbed

C) It must fall toward the chimney so condensate reaches the cleanout

D) It must rise continuously toward the vent, with no dips or sags

Q47. **What is the difference between a service regulator and an appliance regulator?**

A) They are interchangeable, since both do the same job in a gas system

B) Service regulators are indoors; appliance regulators are outdoors

C) Service reduces distribution pressure; appliance reduces it further

D) Appliance regulators handle higher pressures than service regulators

Q48. **What is the maximum horizontal run permitted for a power-vent (Category III) appliance vent?**

A) No limit — power venting can run any distance horizontally

B) As specified by the appliance manufacturer

C) Maximum 3 m regardless of appliance size

D) 600 mm (2 ft) — power vents must be nearly vertical

Q49. **A gas-fired hot water boiler will not fire, and the fault is traced to its low-water cut-off. What condition is that control there to protect against?**

A) The circulating pump losing prime and running dry

B) Return water temperature dropping far enough to condense flue gas

C) Water falling below safe level, which would let the boiler dry-fire

D) Boiler pressure climbing above the relief valve setting on the vessel

Q50. **How do the Red Seal trade descriptions distinguish a Gasfitter Class A from a Gasfitter Class B?**

A) Class A may work above 400,000 Btuh (120 kW); Class B may not

B) Class A covers natural gas only; Class B covers propane only

C) Class A is a regulated trade; Class B is unregulated across Canada

D) Class A is the apprentice level; Class B is the journeyperson level

Answer Key & Explanations

> Scoring: 35 correct out of 50 is 70%. These 50 questions come from the 110-question Gasfitter (Class A) bank; the last page shows what the complete printable bank contains.

Q1. Answer: C

Lockup is the outlet pressure a regulator settles at once downstream demand falls to zero, so it is read with the appliances off, not while one is firing. Some rise above the setpoint is normal, because the seat needs overpressure to close fully, and the measured value on its own therefore neither condemns nor clears the regulator. Acceptability is a comparison: the measured lockup must stay at or below the design pressure of the downstream piping, at or below the maximum inlet pressure marked on the rating plate of every appliance downstream, and within the lockup limit published for that regulator. If it exceeds any one of those, the regulator is readjusted or replaced before the system is left in service. There is no general percentage-of-setpoint rule to fall back on, so take the allowable values from the current CSA B149.1 requirements and the equipment data for the system in front of you.

Q2. Answer: D

In an atmospheric (Bunsen-type) burner: primary air is drawn into the venturi/mixing tube by the venturi effect of gas flow and mixes with gas before combustion at the burner port. Secondary air surrounds the flame from the atmosphere, providing additional O2 for complete combustion. Proper primary/secondary air balance is critical for flame appearance and combustion quality.

Q3. Answer: D

Low-pressure gas systems (residential/commercial) operate at pressures too low for standard pressure gauges. Inches of water column (in WC) — measured with a manometer or magnehelic gauge — is the standard unit. 1 psi = 27.7 in WC. Standard residential NG: 7 in WC. High-pressure systems use kPa or psi.

Q4. Answer: A

The valve that isolates an appliance is a manual quarter-turn valve certified for gas, installed so that it is readily accessible, one for each appliance. Ontario's gaseous fuels Code Adoption Document FS-255-21, which adopts CSA B149.1-20 with Ontario amendments, sets out clause 6.18.2 as requiring a manual shut-off valve in the drop or riser of a residential appliance, or as close as possible to the valve train of a commercial or industrial appliance, or in the horizontal piping between the drop or riser and the valve train, installed so that it is readily accessible; a remote valve for a residential appliance may be up to 50 ft away if it is identified by a metal tag or a permanent sign. Its clause 6.18.4 names the types - ball, eccentric or lubricated-plug - for piping larger than NPS 1, tubing of 1 in OD or larger, or pressures above 0.5 psig. The 2025 edition of CSA B149.1, in its new informative Annex O for industrial occupancy (summarized in Alberta Municipal Affairs' comparison of the 2025 and 2020 editions), adds to clause 6.18.1 that a manual shut-off valve shall be certified to CSA 3.11, CSA 3.16 or CSA/ANSI Z21.15/CSA 9.1, or approved for use with gas, and not be subjected to a temperature or pressure outside its certified rating range; CSA 3.11 covers lever-operated pressure-lubricated plug-type gas shut-off valves, and CSA 3.16 covers lever-operated non-lubricated gas shut-off valves. British Columbia's trades training text says the same in plain words: manual shut-off valves must be of the quarter-turn style, and plug, ball or eccentric types are approved when rated for the pressure and temperature of the service. A ball or lubricated-plug valve certified for gas meets every one of those tests, and its handle reports the state at a glance: in line with the pipe is open, across the pipe is closed. A gate valve fails on each count. It is not one of the named types, it is not certified to the gas shut-off valve standards, and its multi-turn handwheel gives no quick open-or-closed indication. A shut-off valve is not where input is set in any case: input is fixed by the burner orifice and the manifold pressure held by the appliance regulator, not by throttling the supply. A pressure rating alone is not the test either; the valve must be certified or approved for gas. An electrically operated solenoid valve is not a manual valve at all - it opens and closes on a signal from the controls, and the code requires a valve that can be closed by hand regardless of the state of the appliance - and a quick-disconnect device is not accepted as a substitute for the manual valve. 'Readily accessible' has a code definition, reachable quickly for operation, renewal, servicing or inspection without climbing over or removing an obstacle or using a portable ladder, which is why the City of Kelowna requires an additional valve where the only one sits inside a fireplace behind panels that need tools to remove.

Q5. Answer: A

Gas detector placement follows gas behaviour: Natural gas (SG=0.55, lighter than air) rises to the ceiling when leaked — detector within 300mm of ceiling. Propane (SG=1.52, heavier than air) settles to the floor and in low areas — detector within 300mm of floor. Wrong placement means the detector may not alarm until dangerous concentrations exist.

Q6. Answer: B

In a Category III or Category IV system the appliance fan pushes the flue gas, so the inside of the vent sits above the pressure of the room along its whole length. Any gap at a joint therefore pushes combustion products, carbon monoxide included, out into the building. A natural draft vent behaves the opposite way: it runs below room pressure, so a loose joint draws room air inward and dilutes the flue gas instead of releasing it. That difference is why positive-pressure vents are assembled with the listed gaskets, cements or solvent-welded joints plus the mechanical fasteners the vent system calls for, and why the joints are checked for leakage at start-up. Sealing does not change the vent size the sizing tables require, and it is never a substitute for proper support.

Q7. Answer: D

The dip tube is a plastic tube that extends from the cold water inlet at the top of the tank down to near the bottom. Cold incoming water is delivered to the bottom where it can be heated, preventing it from mixing with hot water at the top. Without a dip tube, cold and hot water would mix at the top, reducing efficiency and hot water delivery.

Q8. Answer: B

Clause 6.16.9 of CSA B149.1 (2015 and 2020 editions) reads: 'When piping or tubing passes through masonry or concrete, the portion of piping or tubing that runs through this material shall be sleeved or double wrapped with a pipe wrap tape.' Either method satisfies the clause; a sleeve is not mandatory. Pipe wrap tape is a defined term in the code: adhesive tape of PVC or polyethylene, at least 10 mil (0.3 mm) thick, with a water-resistant adhesive. Where a sleeve is used, Clause 6.16.7 requires it to be of such material and so installed as to protect the piping from damage and galvanic action. Two neighbouring clauses complete the job at a foundation wall. Where the penetration is below ground level, Clause 6.15.9 requires a watertight seal at the point where the piping passes through the outside wall. Where it is above ground, Clause 6.16.8 requires the penetration to be sealed watertight and the portion through the wall to be sleeved or double wrapped. Epoxy paint is not the protection the clause names: painting or coating is the Clause 6.16.1 remedy for piping exposed to a corrosive atmosphere, and it is neither a sleeve nor a double wrap. Fibreglass insulation is not a listed protection at all, and it holds moisture against the steel. Casting the pipe bare in the concrete is what the clause is written to prevent: the pipe is not rebar, and the code gives it its own protection. Edition note: the 2025 edition inserts new sleeve clauses into Clause 6.16 and renumbers the section, and its new Clause 6.16.9 is a different and stricter requirement for piping that passes through an exterior wall to an unheated, inaccessible building element, which must be enclosed in a sleeve, sealed watertight and double wrapped. Read the clause under its number in the edition adopted where you work.

Q9. Answer: B

Air thins with elevation, so a burner fed its sea-level input at 1,000 m is putting the same fuel into less oxygen, and a buoyancy-vented appliance loses draft as well. CSA B149.1 deals with this in Clause 4.22, which Alberta Municipal Affairs reproduces in its STANDATA bulletin on high-altitude installations: for installations between 600 m and 1,350 m above sea level the appliance must be certified for high altitude to CSA 2.17 and adjusted to the high-altitude rating shown on its nameplate, following the manufacturer's instructions. What that adjustment is depends on the model: a smaller orifice, a lower manifold pressure on an appliance that carries its own regulator, a certified field conversion kit, or an air-supply adjustment on an induced-draft or direct-vent unit. The appliance is then marked with its altitude-adjusted input. The adjustment brings the input down, not up, so a larger orifice is the wrong direction and over-fires the burner. The appliance regulator cannot do the job on its own: it holds a gas pressure and has no way of sensing air density. And the utility sets its service pressure from its own distribution system, not from a customer's elevation. Above the highest elevation option the manufacturer provides, the same clause has the input reduced further following the manufacturer's instructions, and only where the manufacturer gives none does a fixed reduction of 4% for each additional 300 m apply. Prove the result with a combustion analysis and label the appliance with its adjusted rating.

Q10. Answer: A

A flame lifts when the velocity of the air and gas mixture leaving the port is higher than the flame speed of that mixture, so the flame cannot anchor on the port and floats above it. Two things raise port velocity: primary air opened too far at the shutter, and over-firing from manifold pressure set above the value on the rating plate. Lifting flames burn unstably, quench against cooler surfaces and make carbon monoxide, and they can blow out and give a delayed ignition on the next call for heat. The correction is to set manifold pressure to the rating plate with a manometer first, then close the primary air until the flame seats on the ports with a clean blue inner cone, and confirm the result with a combustion analysis. The opposite faults look different: too little primary air gives lazy yellow tips and soot, while a restricted orifice or low manifold pressure gives short flames that can flash back into the mixing tube.

Q11. Answer: B

Condensing boilers are designed to operate at low return water temperatures (below 55°C/130°F). At low temperatures, the flue gases cool below their dew point and condense, releasing latent heat. This is the condensing effect — extracting extra energy from the fuel, achieving the highest efficiency (90%+ AFUE). High return temps prevent condensing and reduce efficiency to non-condensing levels.

Q12. Answer: C

An unvented infrared heater discharges its combustion products into the room it heats, so the mechanical ventilation is not a comfort measure, it is the only thing removing those products. The code sizes that ventilation on burner input, requires it to be located so the products from each heater are actually carried outdoors, and holds the space to a ceiling on carbon dioxide measured above the work area. Because the ventilation is the safety device, the code will not let the heater keep firing without it: the system must be interlocked so that any reduction in the airflow required for a heater, or for a group of heaters, shuts that heater or group down. The near miss is the carbon monoxide alarm interlock. An alarm reacts after products of combustion have already built up in the breathing zone, and it cannot see a fan belt that has slipped or a damper that has closed while the burner is still clean; the airflow interlock removes the ignition source at the moment the removal path fails. Running the fan whenever the building is occupied proves nothing about the air it is actually moving, and a general air change rate is not how this ventilation is sized. Watch the other limits on unvented infrared heaters too, including the prohibition on installing them in a residential, care or detention occupancy building.

Q13. Answer: A

CSA B149.1 sets vent terminal height by two separate tests, and both have to be satisfied. First, the terminal must clear the point where the vent passes through the roof by the minimum the code gives for that roof's slope — the flatter the roof the smaller that distance, the steeper the roof the greater it becomes, which is why a flat roof has a low minimum rather than no minimum. Second, the terminal must rise above any portion of the building standing within the horizontal distance the code specifies — a parapet, a penthouse, an adjacent higher wall. A terminal that satisfies only the first test can still sit inside the wind eddy on the lee side of a taller structure, and that eddy is what drags flue gas back down into air intakes, windows and the building itself. The figure for a given job is read out of the code for the roof slope and the surrounding structures actually present, not assumed from a single remembered number, and the manufacturer's certified instructions may impose more.

Q14. Answer: B

The belief that propane, or the odourant in it, rules copper out is shop lore, not the Canadian rule. Clause 6.2.4 of CSA B149.1 accepts copper tubing of Type G (ASTM B837) or Types K and L (ASTM B88) for gas systems with no propane exclusion, and the 2020 edition added Clause 6.2.5, which Alberta Municipal Affairs reproduces in its three-column comparison of that edition: 'Copper tubing Types K and L specified in Clause 6.2.4 b) may be used for liquid propane or propane in the vapour phase.' Alberta's rationale column adds that under ASTM B88 both types are approved for underground and above-ground use. So copper is not confined to natural gas, it is not confined to vapour downstream of the second-stage regulator, and there is no split that reserves Type K for liquid and Type L for vapour: either type may carry either phase. What actually governs a copper propane run is the rest of the code. Joints in seamless copper are made by a flare joint, an approved fitting other than a metallic ball-sleeve compression fitting, brazing with a filler whose melting point exceeds 525°C (1000°F), or a press-connect fitting (Clause 6.9.9). Copper run underground must be Type L or G with an external polyethylene or PVC coating applied at manufacture, or Type K, with any portion above ground protected against physical damage (Clause 6.2.7 in the 2015 edition, renumbered 6.2.8 in 2020). Where it passes through masonry or concrete it is sleeved or double wrapped. Provinces add their own conditions: Saskatchewan's Codes of Practice, for example, allows underground copper to be joined below grade only by brazing and requires a polyethylene shrink-wrap sleeve over the connection on Type L coated liquid-propane tubing. Read the tubing against the code edition and amendments adopted where you work.

Q15. Answer: B

NPC 2020 Sentence 2.2.10.7.(4) states that the temperature of water discharging from a shower head or into a bathtub shall not exceed 49 degrees C. Storing at 60 degrees C is a sound practice for a storage tank, but the code controls what reaches the bather, so the installation needs a temperature control device: the automatic compensating valve required by Sentence 2.2.10.7.(1), or one of the permitted alternatives in Sentences (2) and (3), such as a temperature-limiting device on a tub-only supply or a single tempered water line under an automatic compensating valve. The 43 degree C figure is real but belongs to Sentence (5), which applies to health care facilities and seniors' residences and requires the adjustment at the shower or bathtub controls, so it does not apply to an ordinary dwelling unit. The option to leave the installation alone ignores the fixture limit entirely, and the option to turn the tank down misreads Article 2.6.1.12., which sets a 60 degree C thermostat setting for electric storage-type service water heaters; Note A-2.6.1.12.(1) explains that it is written for electric tanks, which stratify and carry graduated thermostat markings, so it neither caps nor governs a gas water heater.

Q16. Answer: C

The fixed maximum liquid level gauge is a small dip tube whose open end sits at the highest level the liquid is allowed to reach. While the liquid is below that level the bleed valve passes vapour, which flashes to a white haze; the instant the liquid rises to the end of the tube the discharge turns to a liquid spray, and that change is the signal to stop filling. Technical Safety BC's bulletin on cylinder filling gives the two permitted ways of establishing filling density: by weight, to a maximum of 42% of the container's water weight capacity, or by volume, to a maximum of 80% of volume capacity when the container is equipped with a fixed liquid level gauge designed for it. The vapour space left above the liquid is not spare capacity: liquid propane expands sharply as it warms, and a container filled liquid full has nowhere for that expansion to go, so it lifts its relief valve or overpressures the piping it feeds. The near miss is pressure. Container pressure at the end of a fill is simply the vapour pressure of propane at that temperature, and it reads much the same whether the container is a quarter full or nearly full, which is exactly why level has to be established some other way. Nothing about the gauge tells you the relief valve has reseated, and the same bulletin warns that a float gauge or a dispensing meter must not be used to determine filling density.

Q17. Answer: B

The inducer fan draws combustion gases through the heat exchanger and forces them out the vent pipe. It creates negative pressure in the heat exchanger, ensuring complete flue gas evacuation and allowing use of smaller-diameter vent pipes. A pressure switch verifies inducer is running before allowing ignition.

Q18. Answer: C

The clause listing where a vent shall not terminate includes 'less than 7 ft (2.1 m) above a paved sidewalk or a paved driveway that is located on public property'. The height is there to keep the discharge above the people using the walk: flue gas leaves the terminal warm and saturated, and at Canadian winter temperatures it drops its moisture out as frost and glaze on whatever is below it. Terminating high enough puts the plume above head height and keeps the ice off the pavement that the public walks on. The 6 ft (1.8 m) choice is a real figure from the same clause but a different item, the separation from a mechanical air-supply inlet to a building, and that is a separation between two openings, not a height above a walking surface. Two companion figures travel with this one and are worth carrying together: no vent may terminate less than 1 ft (300 mm) above grade level, and no vent may terminate where it could cause hazardous frost or ice accumulations on adjacent property surfaces, which applies whether the surface is public or private.

Q19. Answer: C

CSA B149.1 sorts vented appliances into Categories I to IV by two things: whether the appliance operates with a non-positive or a positive vent static pressure, and whether its flue loss is at least 17% or less than 17%. Category I: non-positive vent static pressure, flue loss not less than 17% - draft-hood-equipped appliances, appliances labelled Category I and fan-assisted appliances for venting into Type B vents. Category II: non-positive, flue loss less than 17%. Category III: positive vent static pressure, flue loss not less than 17%. Category IV: positive, flue loss less than 17%. Saskatchewan's regulator, for example, describes Category III venting as a positive pressure, non-condensing application and refers to Category IV boilers as condensing boilers. A positive vent pressure means the vent must be sealed, with no path for exhaust to escape except at the terminal, so vent material requirements differ for each category. All fuel-burning appliances in Canada are classified as part of certification; pipe material, input rating and the gas burned do not define the category.

Q20. Answer: D

The appliances need at least 5 in WC at their inlets and the system delivers 4 in WC at peak, so either too much pressure is being lost in the run or the supplied pressure is too low to begin with. Both are corrected on the supply side: increase the service pipe diameter so the drop over the 300 m falls, or arrange with the utility for a higher delivered pressure and regulate down at the building. A regulator adds no energy to the gas — it can only reduce and control a pressure that is already there — so a booster regulator has nothing to boost when the upstream pressure itself is inadequate. Opening up the orifices is worse than useless: orifice size is matched to the appliance and to its manifold pressure, and enlarging it to chase a low supply produces an over-fired, badly burning appliance outside its certification. Living with 4 in WC is not an option either, because an appliance below its rated inlet pressure under-fires, burns poorly and can produce CO.

Q21. Answer: D

Natural gas explosive range: LEL = 5% (minimum concentration at which gas will ignite if an ignition source is present) to UEL = 15% (maximum ignition concentration). Below 5% = too lean to ignite. Above 15% = too rich to ignite (but becomes explosive as it disperses). Gas detectors typically alarm at 10-25% of LEL (~0.5-1.25% actual concentration) to provide warning well before reaching the explosive range.

Q22. Answer: A

Written for air rather than for pure oxygen, the reaction is CH4 + 2O2 + 8N2 giving CO2 + 2H2O + 8N2. One volume of methane needs two volumes of oxygen, and because air is only about one fifth oxygen those two volumes arrive escorted by roughly eight volumes of nitrogen, so about ten volumes of air are consumed for every volume of gas. Since a cubic foot of natural gas carries about 1,000 BTU, that is also about 10 cubic feet of air for every 1,000 BTU/h of input, which is the form the figure usually gets used in. Two volumes is the oxygen alone and leaves out the nitrogen that comes with it, so it understates the requirement fivefold. Twenty-five volumes is the propane figure, from C3H8 + 5O2 + 20N2, because the larger molecule needs two and a half times the air - a real number, but for the wrong fuel. Fifteen volumes is the near-miss: it is what a burner actually draws once the roughly 50 percent excess air that real appliances are set up with is added, so it answers a different question than the theoretical requirement, and total air including the dilution air pulled in at a draft hood is higher again.

Q23. Answer: D

Standard white PTFE tape is not rated for fuel gas service. CSA B149.1 Clause 6.9.6 requires that when a jointing sealant is used it shall be certified to CAN/ULC-S642 and shall be applied to the male threads of a metal pipe, and that tape shall be stretched and applied in a clockwise direction with a 50% overlap leaving the first two starter threads bare. Gas-rated PTFE tape is normally yellow, and gas-rated pipe thread compounds carry the same CAN/ULC-S642 certification; a general-purpose sealant chosen off the shelf has no such certification and can soften or wash out in the presence of gas and its liquids, giving a joint that passes the pressure test and weeps months later. Leaving the first two threads bare keeps sealant from being carried into the bore where it can foul a control or an orifice. Tapered NPT threads do not seal on their own either: the thread form leaves a spiral leak path, and filling it is exactly what the sealant is for.

Q24. Answer: B

Black steel pipe with malleable iron or steel fittings is the standard for above-ground gas piping inside a building: it is strong, it threads and welds readily, and it is the material the steel piping requirements of the installation code are written around. Galvanized steel is the same base product — ASTM A53/A53M is titled Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless, so black and galvanized pipe are two finishes of one specification — but galvanized is not the customary material for interior gas distribution and the zinc coating buys nothing in a dry indoor run. PVC is never a fuel gas material anywhere in the system: plastic gas piping in Canada is polyethylene to CSA B137.4, Polyethylene (PE) Piping Systems for Gas Services, and it is a buried outdoor material, not an indoor one. Copper tube is accepted for gas in specified applications and sizes, and it is common on short appliance runs, but it is not the usual choice for distribution piping through a building. Confirm the acceptable materials, the sizes and any restriction against the edition of CSA B149.1 in force in your jurisdiction together with the provincial amendments.

Q25. Answer: C

Clocking the meter is a volume measurement and nothing else. With every other gas appliance off, you time how long the smallest test dial on the meter takes to complete one revolution; that dial is marked in a known volume, so the timing gives cubic feet per hour — the volume of gas the appliance is actually drawing. Input is then a calculation rather than a reading: multiply the measured flow by the heating value of the gas to get BTU per hour, and compare that against the input on the rating plate. Nothing in the test reads pressure. Manifold pressure is read with a manometer at the burner test port and inlet pressure at the appliance inlet port, which is a separate check. Nothing in it reads heating value either — that is a property of the gas the utility supplies, and it has to be obtained from the utility if the result is to be accurate. And efficiency is not measured here at all: clocking says how much fuel goes in, not how much of it becomes useful heat, which takes a combustion analysis.

Q26. Answer: C

A drip leg is a short vertical pipe extension below the horizontal run, capped at the bottom, installed as the last fitting before connecting to the appliance. Condensate, pipe scale, and debris fall into the drip leg before they can enter the appliance regulator and gas valve, preventing blockage and damage.

Q27. Answer: B

Natural gas combustion produces CO2 and H2O. The dew point of natural gas flue gases is approximately 55-60°C (130-140°F); below this temperature, water vapour condenses into liquid in the vent. The condensate absorbs CO2 forming carbonic acid (H2CO3) and sulfurous acid, producing an acidic liquid (pH 3-4). This is why ordinary metal vents corrode when serving Category IV condensing appliances.

Q28. Answer: A

A spillage test verifies combustion gases are actually going up the vent and not spilling into the room. It must be performed after each installation and commissioning of a natural draft appliance. Method: hold a smoke pencil or incense stick at the draft hood opening — smoke should be drawn up into the vent (not spill into room). Test with all exhaust fans running (worst case). Spillage indicates blocked vent, insufficient combustion air, negative pressure in building, or improper vent sizing.

Q29. Answer: B

A non-recirculating direct gas-fired industrial air heater has no heat exchanger and no vent. The burner fires straight into the air stream, all of its supply air is ducted from outdoors, and the combustion products are delivered into the space with the heated air, which is why the code confines these heaters to industrial buildings and forbids them to supply air to an area where sleeping accommodation is provided. Sleeping occupants cannot be relied on to notice a problem and leave, so no dilution rate is accepted as adequate for them. The near miss is the assumption that continuous occupancy is the disqualifier. It is not: a machine shop running three shifts is exactly the kind of industrial space these heaters are built for, and so are a racked warehouse and a loading dock cycling its doors in winter, where the heater's outdoor air also replaces what the doors blow out. Class A work brings these heaters in regularly as make-up air and door heaters. Watch the other limits with them, including clearance to combustible material as marked on the rating plate, keeping the heater well away from any plane where combustible gas, vapour or dust is present, and interlocking outside dampers or louvres so the main burners cannot fire until they are fully open.

Q30. Answer: B

The standard low-pressure natural gas service to a house is about 7 inches of water column (7 in WC, 1.75 kPa). That is the outlet setting of the service regulator at the meter set, and it is the pressure delivered into the building piping. It is not the same number at every point downstream. At the appliance inlet the pressure is nominally about 7 in WC less whatever the building piping drops while gas is flowing, which is why inlet pressure is read at the appliance with the appliance firing, and why the rating plate states a minimum and a maximum inlet pressure rather than a single figure. The appliance's own regulator, inside the gas valve, then reduces the inlet pressure to the manifold (burner) pressure, about 3.5 in WC (0.87 kPa) for natural gas, so 3.5 in WC is a burner figure, not a service figure. 14 in WC is twice the service setting, not the delivered service pressure. 2 psig (13.8 kPa) is an elevated-pressure service that a utility may offer for larger loads, with a regulator at the building bringing it back down.

Q31. Answer: A

IPI (Intermittent Pilot Ignition): a spark creates a pilot flame at the start of each heat cycle; the flame sensor confirms pilot is lit, then the main gas valve opens and the pilot ignites the main burner. DSI (Direct Spark Ignition): the spark fires directly at the main burner without a pilot. DSI is faster but requires the spark to be at the burner.

Q32. Answer: B

CSA B149.1 (Natural Gas and Propane Installation Code) is the primary national code governing gas fitting work in Canada. It covers: gas supply systems, piping materials and sizing, appliance installation, venting, combustion air, pressure testing, and safety requirements. Adopted by provinces with local amendments.

Q33. Answer: B

A diaphragm regulator works by balancing gas pressure under the diaphragm against a spring, with the far side of the diaphragm open to atmosphere through the vent. Block that vent and the regulator loses its atmospheric reference and can no longer hold the outlet where it was set. The vent is also the relief path: if the diaphragm splits or the seat fails to close, gas is meant to discharge from the vent to a safe location outdoors. CSA B149.1's definition of a safe location for venting gas, reproduced in Alberta Municipal Affairs' published comparison of the 2020 edition, is one where you can reasonably expect to prevent, among other things, 'blocking of the vent termination by snow, ice, water, or any other object or thing' — so a vent packed with ice is by definition not in a safe condition. TSSA and Enbridge Gas make the same point to homeowners: a gas meter covered in snow or ice can impede its function of controlling the pressure of the gas supplied to a home. Clear it gently with a soft brush, never by kicking, hitting or chipping at it. The vent is not a rain drain, it does not feed air into the gas stream, and it has nothing to do with the meter index.

Q34. Answer: D

Terminating under a veranda, porch or deck is conditional, not forbidden, but two conditions have to be satisfied together. The structure has to be fully open on a minimum of two sides beneath the floor, and the distance between the top of the vent termination and the underside has to be greater than 1 ft (300 mm), measured from the top of the vent to the bottom of the joists. Lattice skirting on all four sides fails the first condition outright, so the 400 mm clearance is beside the point: the two conditions are cumulative, and satisfying only the vertical one is what makes the 400 mm answer the trap. The reasoning is simple to picture. An enclosed underfloor space has no cross-ventilation, so the plume has nowhere to go, and combustion products, moisture and frost collect under the deck where children and pets sit. A terminal cap does not create ventilation, and the certified plume-redirecting fitting belongs to a different requirement altogether, the one that governs a Category IV or special-vent terminal discharging into a narrow side yard toward a property line. Open the skirting on at least two sides, or find another wall.

Q35. Answer: A

LP gas systems typically use two-stage regulation: First stage at the tank reduces vapor pressure (which varies with temperature) to a constant ~10 psig (69 kPa). Second stage at or near the building reduces 10 psig to appliance operating pressure (~11 in WC / 2.74 kPa). This provides stable pressure regardless of tank level or temperature.

Q36. Answer: A

Pipeline natural gas is primarily methane (CH4), typically 90-95%, with minor amounts of ethane (C2H6), propane (C3H8), and butane (C4H10), plus inert gases (N2, CO2). The exact composition varies by source and region. This composition determines energy content (~1,000 BTU/ft3) and combustion characteristics.

Q37. Answer: C

Stop what you are doing and operate nothing that can make a spark — no light switches, no cellphone or landline, no lighter, no matches. Leave the building at once, leaving the door open behind you and any window that is already open. Do not stay inside to ventilate the place, and do not stay inside to hunt for the leak with soap solution; both of those keep you in the building while a flammable mixture is building up in it. Shut the gas off at the exterior meter valve only if that valve can be reached safely from outside. From outside, call 911 and the gas utility's 24-hour emergency line. Do not go back in until an official says it is safe.

Q38. Answer: C

Stoichiometric combustion uses exactly the theoretical air-fuel ratio to completely combust all fuel — no excess air and no incomplete combustion: all fuel molecules react with exactly enough oxygen molecules, producing only CO2 and H2O with no excess O2 remaining. It is a calculation, not a setting anyone runs a burner at. In practice gas appliances are set up with roughly 15% to 50% excess air (about 3% to 7.5% O2 in the dry flue gas), power burners at the low end and residential atmospheric appliances at the high end, because mixing is never perfect and any shortfall of air makes carbon monoxide.

Q39. Answer: C

Gas dryer exhaust carries moisture, heat, lint, and combustion products including carbon monoxide. The building code article on venting of laundry-drying equipment requires the exhaust duct or vent connected to that equipment to discharge directly to the outdoors, and requires the duct to be independent of other exhaust ducts, accessible for cleaning, and made of a smooth corrosion-resistant material. Discharging into an attic, crawl space, garage, wall cavity, or any lint trap inside the building creates a lint fire hazard, moisture damage, and a carbon monoxide risk.

Q40. Answer: D

HSI (hot surface ignitor) is made of silicon carbide or silicon nitride. The control board energizes it for a warm-up period (15-30 seconds) until it glows red-hot, then opens the gas valve. The glowing element ignites the gas. HSI replaced standing pilots in mid-efficiency and high-efficiency furnaces.

Q41. Answer: B

Clause 6.16.4 of CSA B149.1 sets the trigger by distance, not by material, pressure or length: 'Tubing run inside hollow walls or partitions within 1.75 in (43 mm) of the surface shall be protected against physical damage and puncture at the joists, studs, and plates by the use of No. 16 USG (1.59 mm) plates or sleeves.' The logic is the fastener. A drywall screw or trim nail driven at a stud or plate reaches only so far, so tubing lying deeper than 43 mm from the surface is out of its path, while tubing lying closer must have steel between it and the fastener. The clause adds that it does not apply to tubing that passes directly through a wall or partition. It applies to tubing of every kind the code recognizes, which Clause 6.2.8 lists as CSST, seamless copper and seamless steel, so copper gets no exemption on account of its wall, and neither the operating pressure nor the length of the run enters into it; the 6 ft (2 m) figure a candidate may recall is the support spacing for tubing in Table 6.2, not a protection trigger. For CSST there is a second layer. Clause 6.16.13 (2015 numbering) requires CSST and its fittings to be protected against physical damage in accordance with the manufacturer's certified installation instructions and with the code, and Alberta's gas safety bulletin G-04-01 repeats that CSST 'shall be protected against physical damage in accordance with the manufacturer's certified installation instructions'. Those instructions can protect a wider zone than the code minimum. Manitoba's gas equipment bulletin ITSM 17-011 states the same 16-gauge plate rule for tubing less than 1 3/4 in from the exposed edge of a stud, joist or plate, and then says that where the manufacturer's installation instructions exceed that minimum, the manufacturer's instructions are to be followed. Concealed tubing must also be inspected and pressure tested in its final position before the wall is closed (Clauses 6.7.1 and 6.22.5).

Q42. Answer: C

The clearance to an openable window or door is banded by appliance input, not fixed. Alberta Municipal Affairs' STANDATA gas safety information bulletin G-01-10, 'Sidewall vent terminations', quotes the code clause in full: a vent shall not terminate within 6 in (150 mm) for inputs up to and including 10 000 Btuh (3 kW); 12 in (300 mm) for inputs from 10 000 Btuh (3 kW) up to and including 100 000 Btuh (30 kW); and 3 ft (900 mm) for inputs exceeding 100 000 Btuh (30 kW). The same distances apply to a nonmechanical air-supply inlet and to the combustion air inlet of another appliance. An input of 60,000 BTU/h sits in the middle band, so 300 mm applies. The 6 ft (1.8 m) figure belongs to a different requirement in the same clause - the clearance to a mechanical air-supply inlet, which draws combustion products in rather than waiting for them to drift.

Q43. Answer: A

Pre-purge exists to sweep the combustion chamber, the heat exchanger passages and the vent with fresh air before any ignition source is energized, so that any fuel that accumulated while the burner was off is diluted and carried out of the appliance. How long that takes is a volume problem: the fan delivers a known airflow at purge rate, the chamber and flue hold a known volume, and the purge must run long enough to deliver the number of air changes specified for that burner in the manufacturer's burner management sequence. A larger chamber, a longer flue, or a lower purge-rate airflow all lengthen the required pre-purge. The other three answers name real steps in a light-off sequence, but none of them is what the purge timer is sized from: ignitor warm-up, refractory warm-up and the flame-proving period are separate, independently timed events. Post-purge works the same way at the other end of the cycle, clearing residual products of combustion after shutdown.

Q44. Answer: D

AFUE of 80% means 80% of fuel energy is converted to heat delivered to the space. Output = Input × AFUE = 100,000 × 0.80 = 80,000 BTU/h. The remaining 20,000 BTU/h (20%) is a seasonal-average loss: AFUE is a seasonal efficiency, so that loss includes not only heat carried away in the combustion gases but also losses from cycling on and off, cold start-ups and warm house air drawn up the chimney.

Q45. Answer: B

Direct-vent (sealed combustion) appliances use a two-pipe system: one pipe draws combustion air from outside, and the other exhausts flue gases to outside. The combustion process is completely isolated from the building interior, making them ideal for tight buildings and eliminating combustion air requirements.

Q46. Answer: D

A natural draft appliance has nothing but buoyancy to move its flue gases, so the connector must work with that buoyancy: it rises continuously from the appliance outlet to the chimney, is kept as short as the installation allows, and carries no dip or sag anywhere along its length. A sag collects condensate and becomes a partial blockage that pushes products of combustion back out of the draft hood. A level run gives the gases no help at all, and a run that falls toward the chimney drains condensate along the connector seams and stalls the draft on start-up, when the flue is still cold and the buoyancy is weakest. Numeric pitch figures in common circulation come from United States practice and from vent manufacturers' literature; in Canada the certified instructions for the appliance and the vent system govern the installation.

Q47. Answer: C

The two regulators sit in series and each takes the pressure down one step. The service regulator at the meter set reduces the utility's distribution pressure to the service pressure delivered into the building, about 7 in WC (1.75 kPa) for natural gas. By the time the gas reaches an appliance it has lost a little more in the building piping under flow, so the appliance inlet pressure is nominally about 7 in WC and has to sit within the minimum and maximum marked on the rating plate. The appliance regulator, built into the combination gas valve or fitted on the supply, then takes that inlet pressure down to the manifold (burner) pressure the manufacturer specifies: about 3.5 in WC for natural gas and about 10 in WC for propane. Inlet and manifold pressure are measured at two different points and are never the same number, because a regulator can only reduce a pressure that is already there. So the appliance regulator handles the lower of the two pressures, not the higher; the service regulator sits at the meter set, normally outdoors, and the appliance regulator sits at or inside the appliance, so the indoor-outdoor claim is backwards; and the two are not interchangeable, since each is sized and sprung for its own inlet and outlet range. On propane the second-stage regulator at the building plays the service role, delivering about 11 in WC (2.74 kPa) to the appliance inlet.

Q48. Answer: B

Power-vent (Category III) appliances use a blower to push flue gases through the vent. Maximum horizontal run depends on the appliance manufacturer's specifications and the vent system's static pressure capacity — the certified installation instructions list the allowable vent length for each pipe size and number of elbows, and those limits differ widely from one model to another. Each elbow reduces equivalent straight-pipe capacity. Exceeding maximum run reduces flue gas velocity and can cause condensate accumulation.

Q49. Answer: C

A low-water cut-off senses the water level in the boiler and interrupts fuel to the burner when the level falls below the safe minimum. Firing a boiler whose water level has dropped below the heated surfaces overheats the metal, which can crack the vessel or heat exchanger; on a steam boiler, feeding cold water back onto surfaces that have been fired dry can produce a violent release. The cut-off does not restore the water by itself unless it is paired with a feeder, so a low-water lockout is a signal to find the leak or the feed fault before resetting, not simply to reset and walk away. Each of the wrong answers belongs to a different control the same boiler may carry: loss of circulation is the flow switch's job, overpressure is the relief valve's job, and low return water temperature is handled by a boiler-protection or low-limit aquastat arrangement. Sorting these four apart is the point - they are all 'safety limits' but they sense different things.

Q50. Answer: A

The Red Seal trade descriptions draw the line at appliance and equipment input rating. Gasfitters Class A work on appliances and equipment including those exceeding 400,000 Btuh (120 kW), which brings in boilers, burners, makeup air units, process burners and similar large equipment; Gasfitters Class B work on fuel-fired appliances that do not exceed that rating, in residential, institutional and commercial buildings and restaurants. The two are separate certifications rather than apprentice and journeyperson stages of one ladder, neither is defined by a single fuel since both work with the fuel gases their jurisdiction licenses them for, and gas fitting is a regulated trade across Canada in either class. Licence titles, classes and the exact scope each one carries are set by the provincial or territorial gas authority, so confirm local scope before working outside the class you hold.

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